

Spatially-resolved transcriptomic data integration

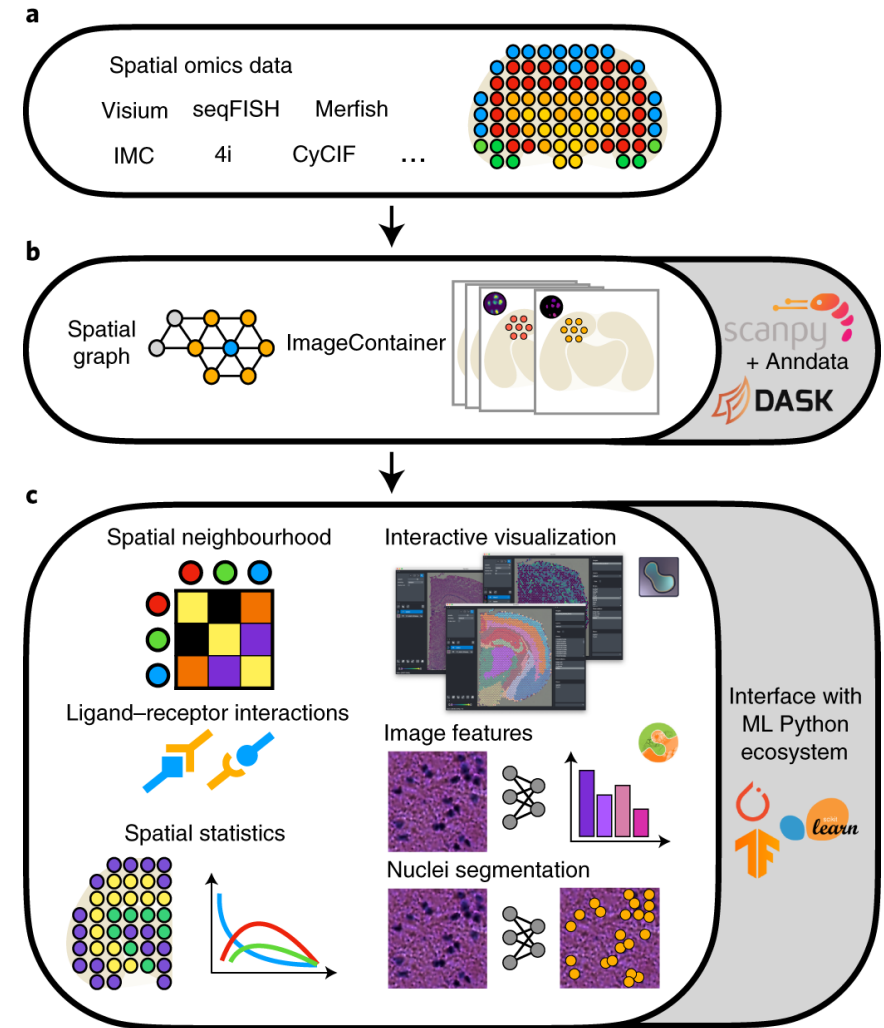
Ahmed Mahfouz

Department of Human Genetics, Leiden University Medical Center
Pattern Recognition and Bioinformatics, TU Delft

Spatial data analysis pipelines

- Image preprocessing
- Cell segmentation / segmentation-free
- Quality control / filtering / normalization
- Downstream analysis:
 - Cell annotation
 - Spatially-aware clustering (aka. Domain identification)
 - Cell-cell communication
 - Deconvolution
 - Predicting unmeasured genes
 - Consecutive sections alignment
 - ...

Pipelines: Squidpy [Palla,...Theis, 2022], SPARROW [Pollaris*, Vanneste*,...,Saeys 2024], MCMICRO [Shapiro,..., Sorger 2021], SOPA [Balmpey,..., Cournède 2024],...



Squidpy: Palla,...Theis (2022)

Integration challenges in SRT

- Integrating SRT and sc/snRNA-seq data
- SRT data alignment (pseudo 3D, virtual block,...)
- Spatial multiomics

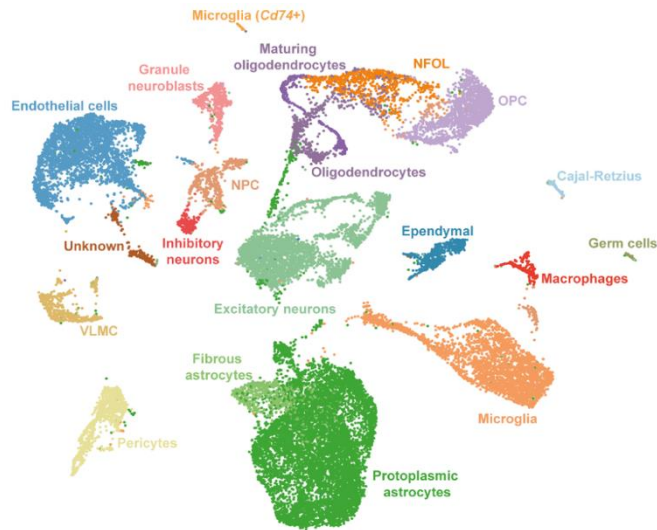
Integration challenges in SRT

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Different technologies -> different questions

Single-cell/nuclei RNA-seq

(droplet, combinatorial indexing, plate-based)

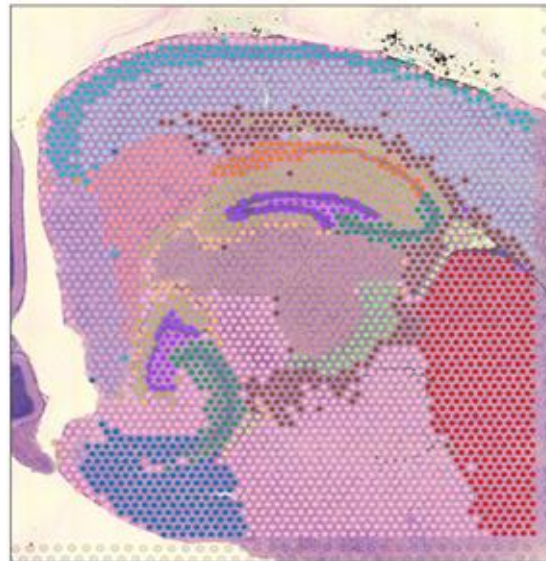


+ “Untargeted”

- No spatial information

NGS-based

(Visium, Slide-seq, Stereo-seq, Open-ST,...)

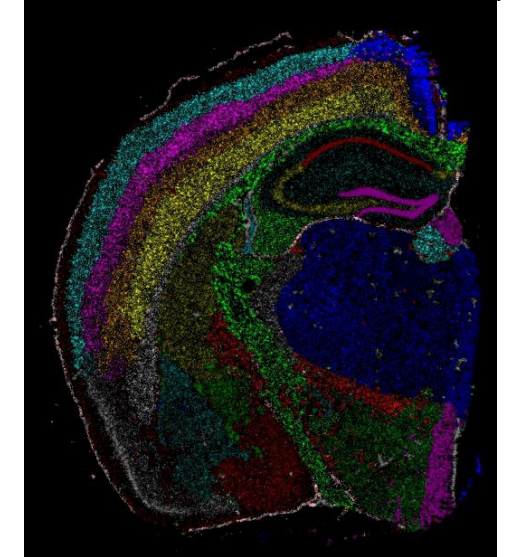


+ “Untargeted”

- Low resolution (spots)

In-situ sequencing / x-FISH

(Xenium, EEL-FISH, MERFISH, seqFISH)

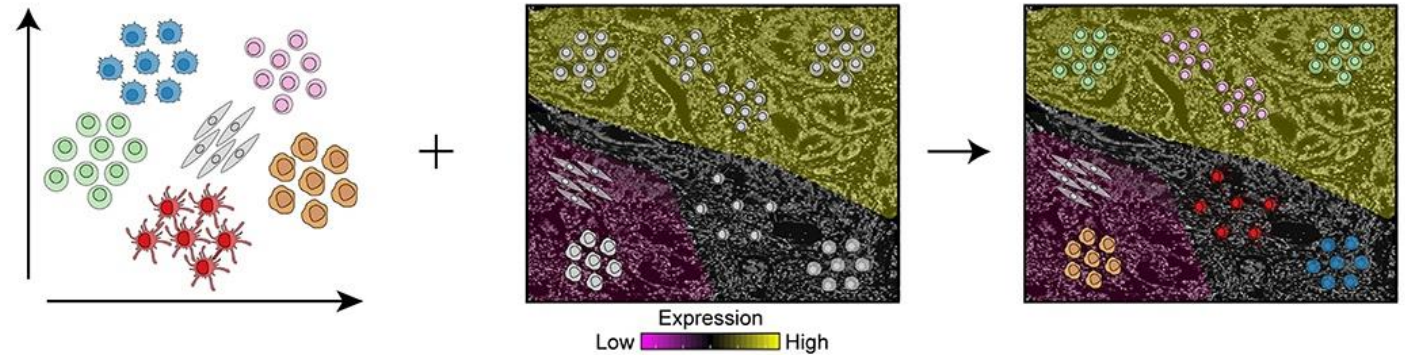


+ High resolution

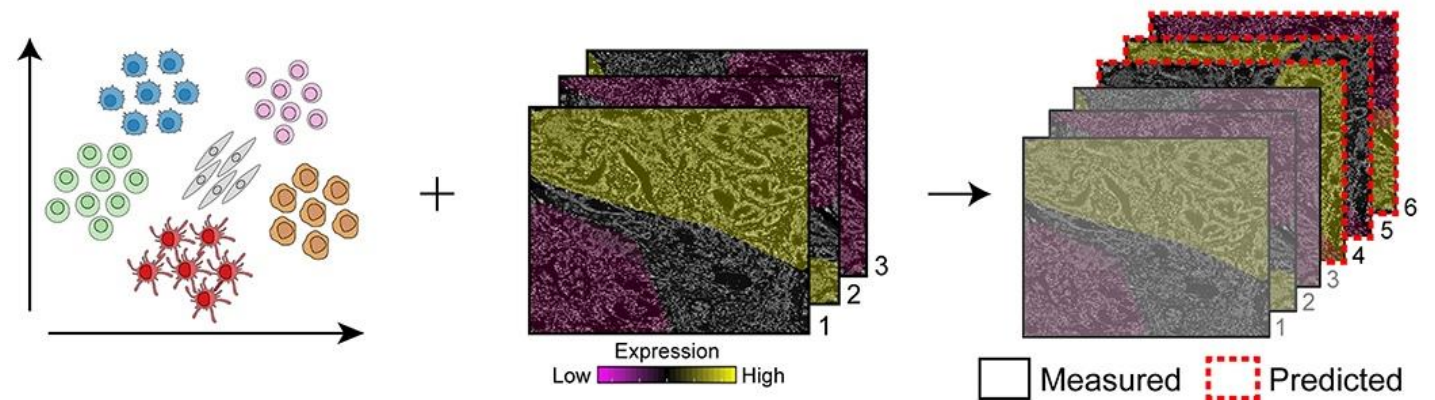
- limited genomic features

Single-cell + spatial

- Predict the location of dissociated cells
- Deconvolution



- Predicting spatial expression of unmeasured genes



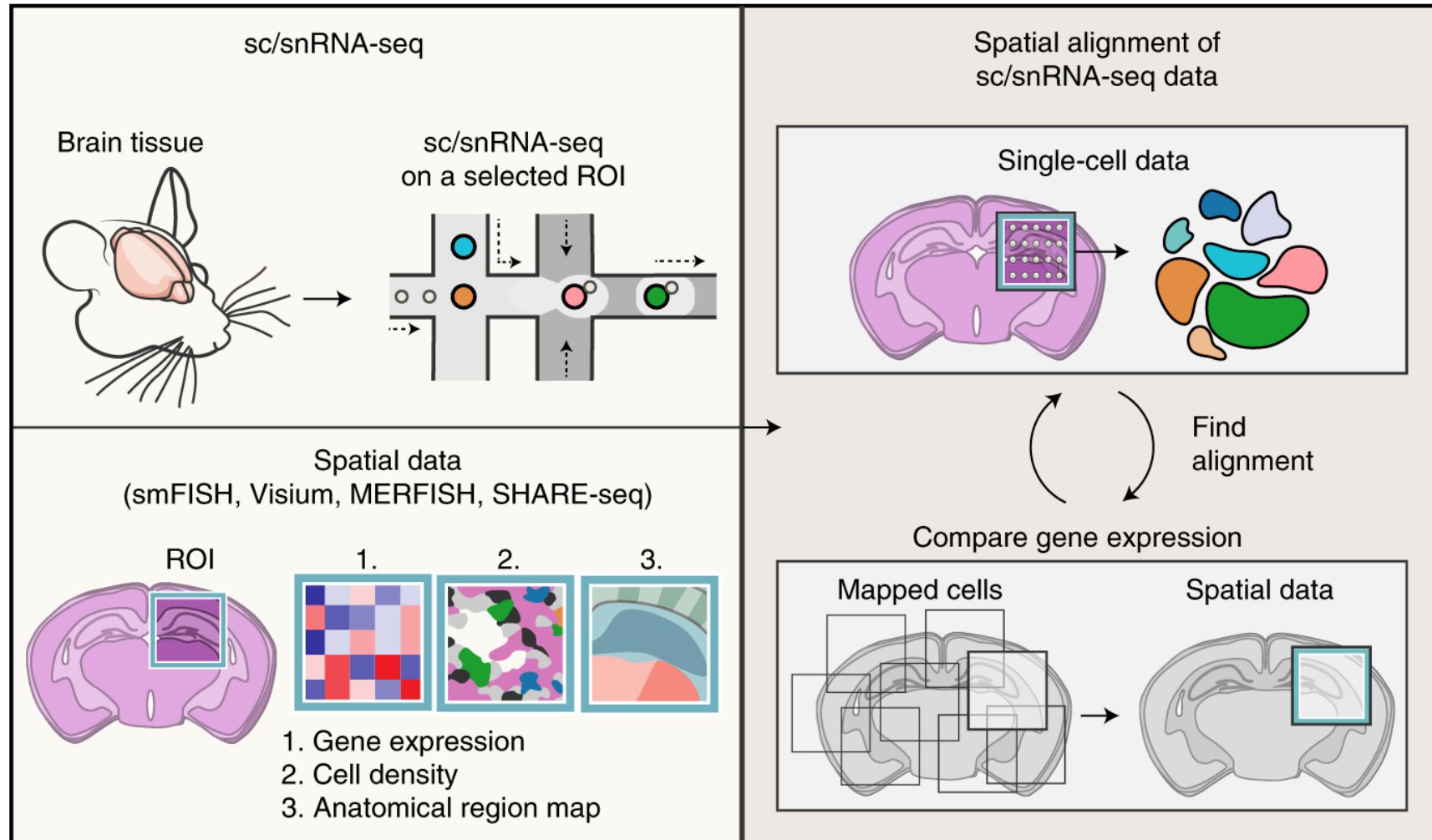
Approaches for scRNA-seq & spatial data integration

	Localization	Deconvolution	Imputation
Joint embedding e.g. SpaGE, Seurat, LIGER, gimVI, ENVI, stPlus DSTG, SD			
Probabilistic modelling e.g. CARD, cell2location, RCTD, stereoscope, SpatialDecon, STRIDE, NMFreg, SpatialDWLS, SPOTlight, Stdeconvolve, SpiceMix, Berglund			
Probabilistic mapping e.g. Tangram, novoSpaRc, SpaOTsc			

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Tangram



Tangram

- Given scRNAseq ($S_{n_{cells} \times n_{genes}}$), and spatial data ($G_{n_{voxels} \times n_{genes}}$)
- Learn a mapping matrix $M_{n_{cells} \times n_{voxels}}$, $M_{ij} \geq 0$ is the probability of cell i of being in voxel j , $\sum_j^{n_{voxels}} M_{ij} = 1$.
- $M^T S$: predicted spatial gene expression, $m_j = \sum_i^{n_{cells}} \frac{M_{ij}}{n_{cells}}$: predicted cell density in voxel j

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- Objective function:

$$\Phi(\tilde{M}) = \underbrace{KL(\vec{m}, \vec{d})}_{\text{Density term}} - \underbrace{\sum_k^{n_{genes}} \cos_{sim}((M^T S)_{*,k}, G_{*,k})}_{\text{gene/voxel expression term}} - \underbrace{\sum_k^{n_{voxels}} \cos_{sim}((M^T S)_{j,*}, G_{j,*})}_{\text{voxel/gene expression term}}$$

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Tangram

- Actual objective function:

$$\begin{aligned}\Phi \left(\tilde{M}, \vec{\tilde{f}} \right) = & KL \left(\vec{\mathbf{m}}^f, \vec{\mathbf{d}} \right) - \sum_k^{n_{genes}} \cos_{sim} \left((M^T S^f)_{*,k}, G_{*,k} \right) \\ & - \sum_j^{n_{voxels}} \cos_{sim} \left((M^T S^f)_{j,*}, G_{j,*} \right) - \lambda_{r_1} \sum_{i,j}^{n_{cells}, n_{voxels}} M_{ij} \log (M_{ij}) \\ & + abs \left(\sum_i^{n_{cells}} f_i - n_{target_cells} \right) + \sum_i^{n_{cells}} (f_i - \tilde{f}_i^2).\end{aligned}$$

Tangram

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Tangram

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$\vec{f_{n_{cells}}}$: filter vector

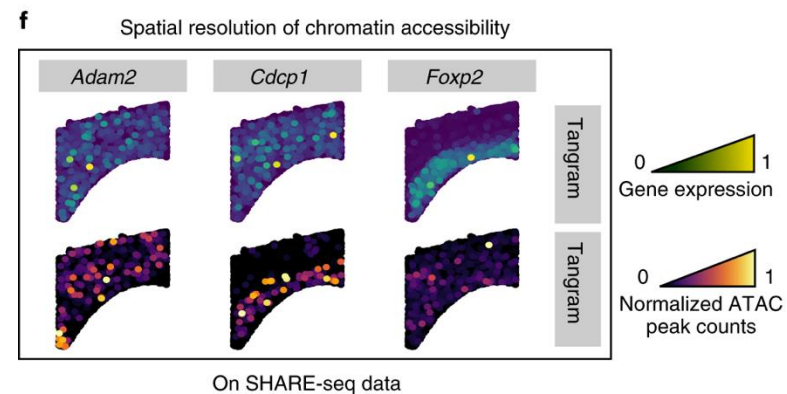
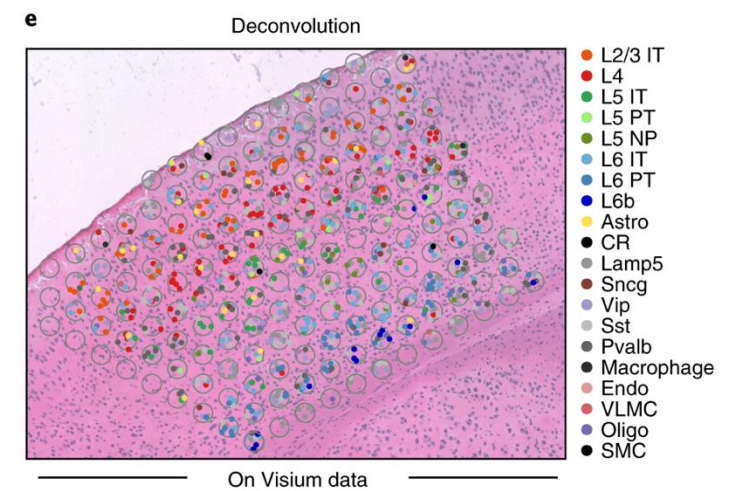
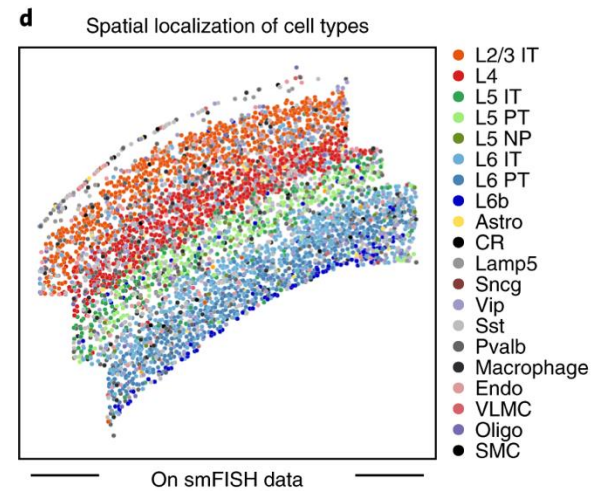
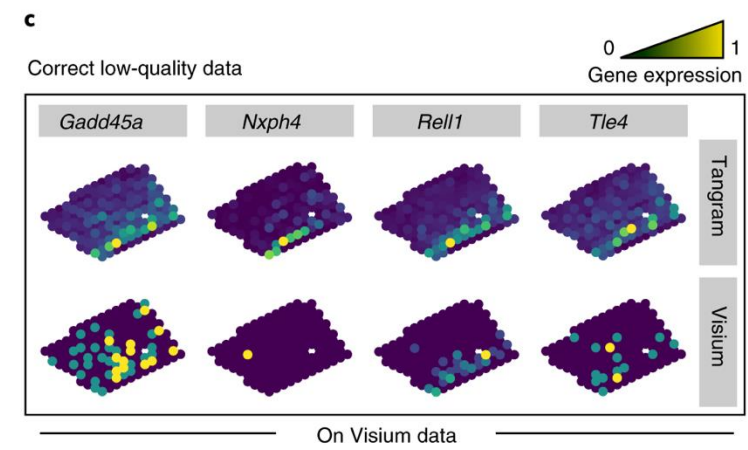
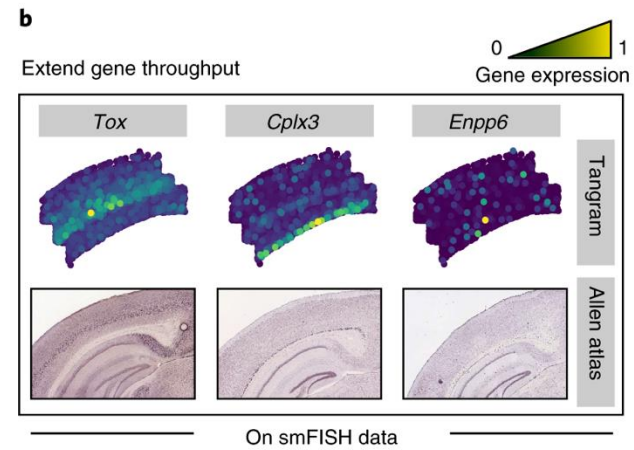
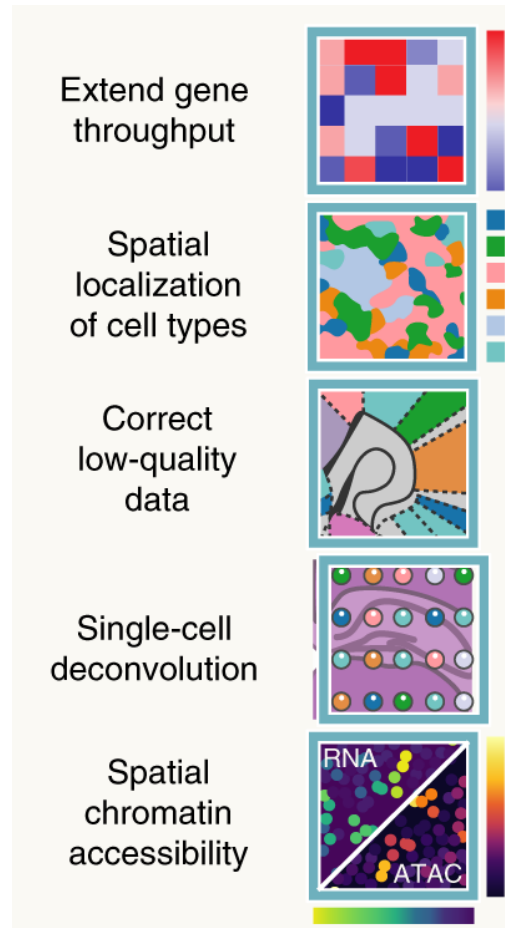
$$- \sum_j^{n_{voxels}} \cos_{sim}((M^T S^f)_{j,*}, G_{j,*}) - \lambda_{r_1} \sum_{i,j}^{n_{cells}, n_{voxels}} M_{ij} \log(M_{ij})$$

$$+ abs(\sum_i^{n_{cells}} f_i - n_{target_cells}) + \sum_i^{n_{cells}} (f_i - f_i^2).$$

Count term

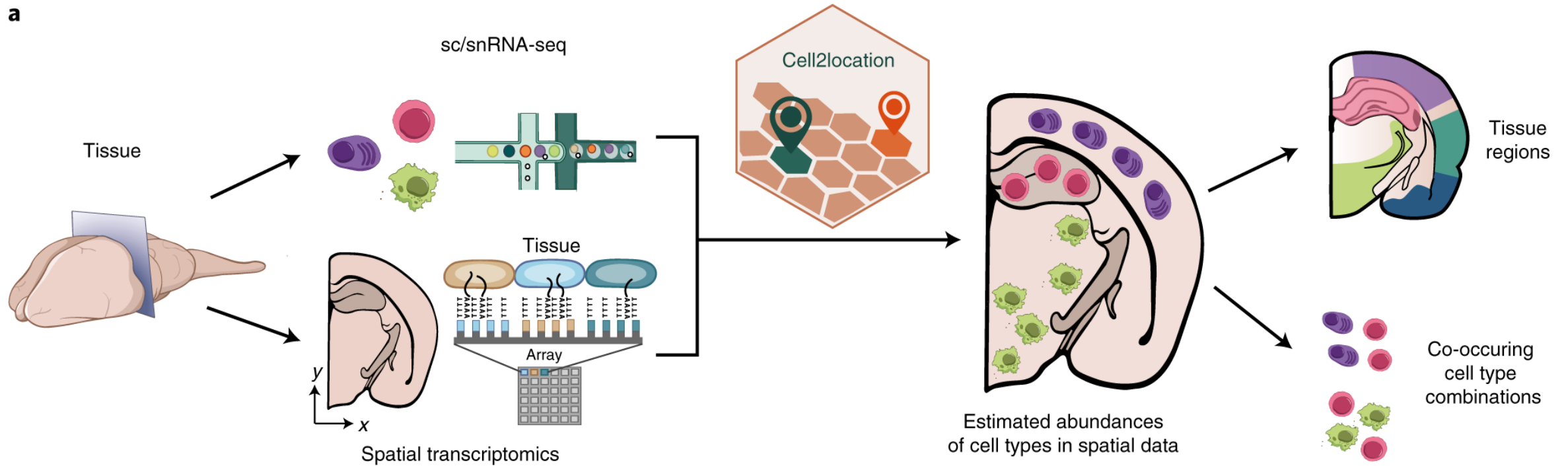
Filter regularizer
(promote Boolean values)

Applications of Tangram



cell2location

a



cell2location

Spatial counts $\sim NB(\mu_{s,g}, \alpha_{e,g})$

s: spatial location

g: gene

e: batch (section, slide, ...)

cell2location

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$$\mu_{s,g} = \left(\underbrace{m_g}_{\text{technology sensitivity}} \cdot \underbrace{\sum_f w_{s,f} g_{f,g}}_{\text{cell type contributions}} + \underbrace{s_{e,g}}_{\text{additive shift}} \right) \cdot \underbrace{y_s}_{\text{per-location sensitivity}}$$

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$$\mu_{s,g} = \left(\underbrace{m_g}_{\text{technology sensitivity}} \cdot \underbrace{\sum_f \underbrace{w_{s,f} g_{f,g}}_{\text{cell type contributions}}}_{\text{Reference signature from scRNA-seq data}} + \underbrace{s_{e,g}}_{\text{additive shift}} \right) \cdot \underbrace{y_s}_{\text{per-location sensitivity}}$$

Cell type abundance.
Prior: similarity in cell
type between
locations

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Reference signature from scRNA-seq data

To account for differences between scRNA-seq and spatial

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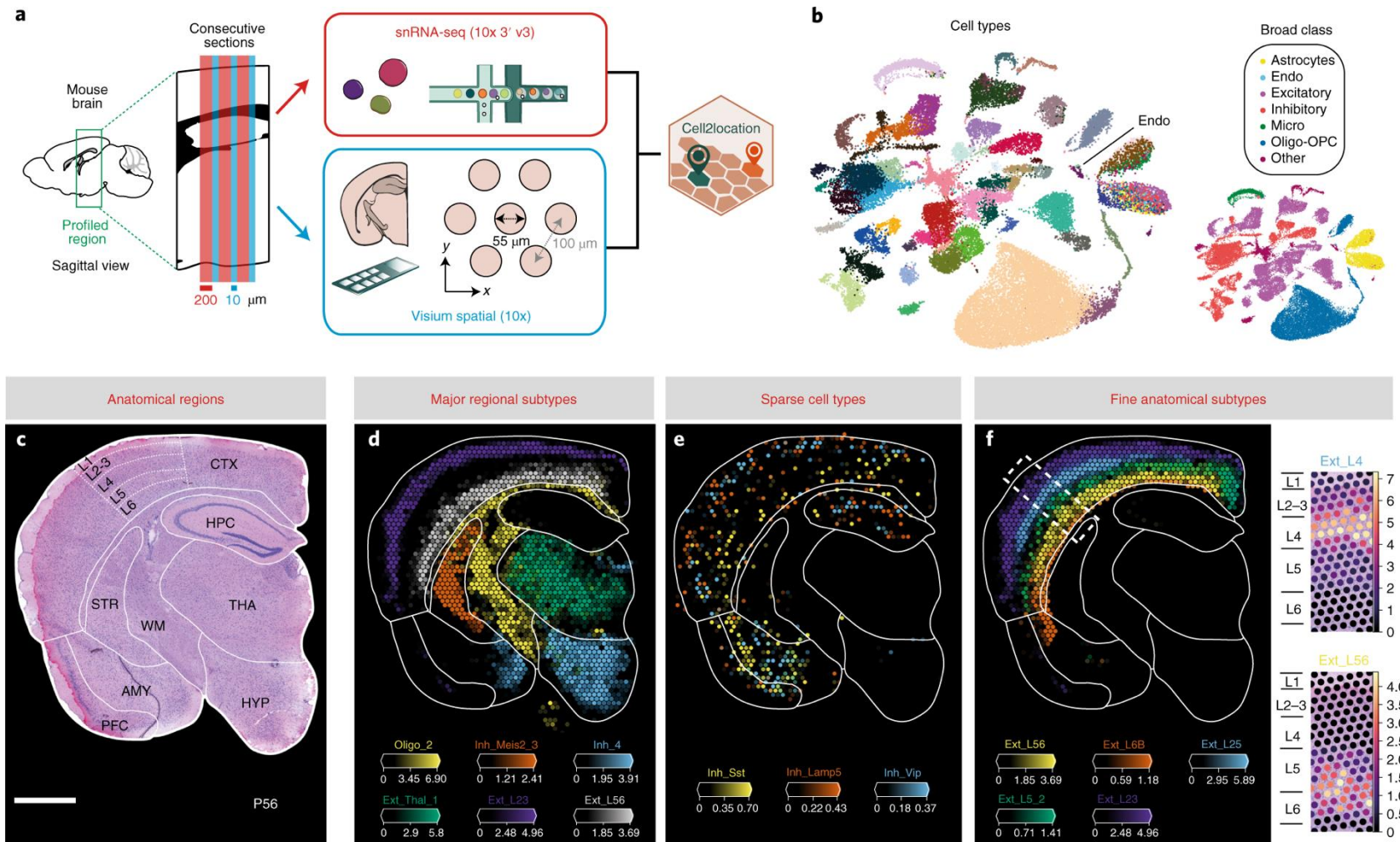
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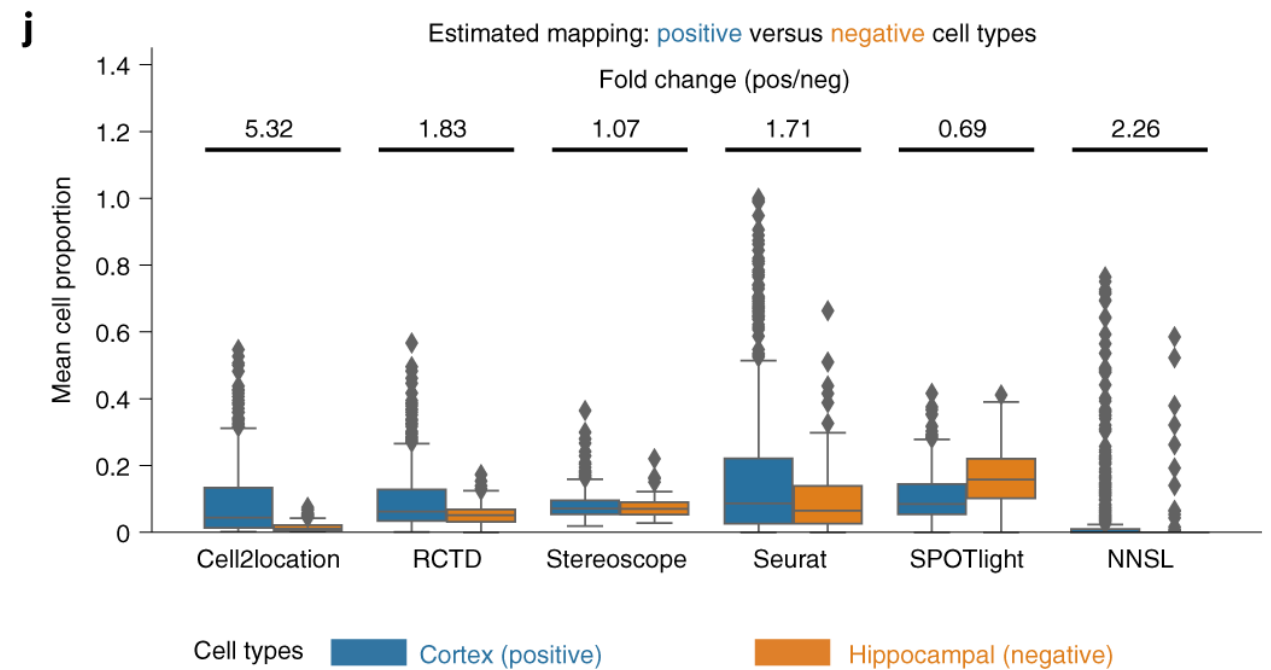
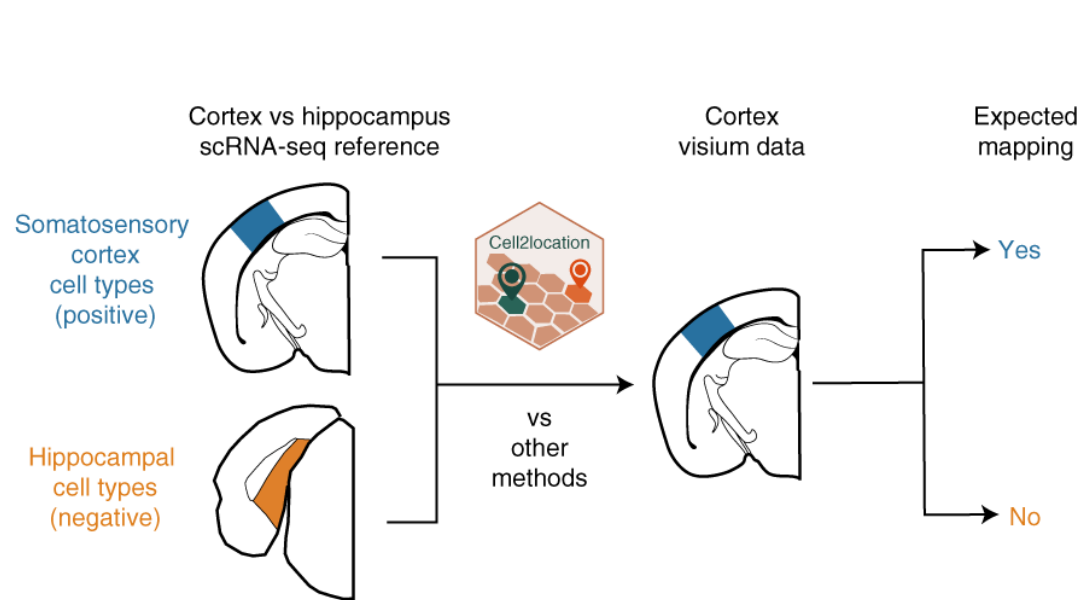
To account for free floating RNA (background noise)

To account for differences in sensitivity within a section

Application of cell2location



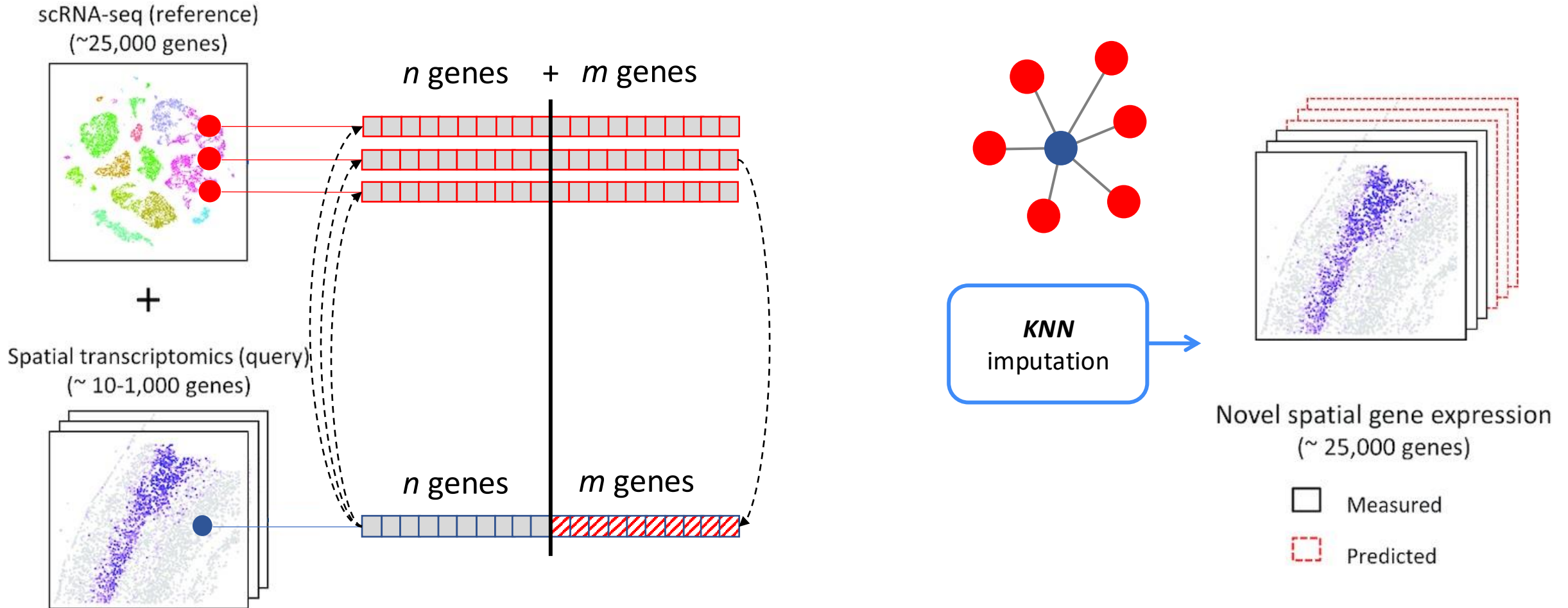
Application of cell2location





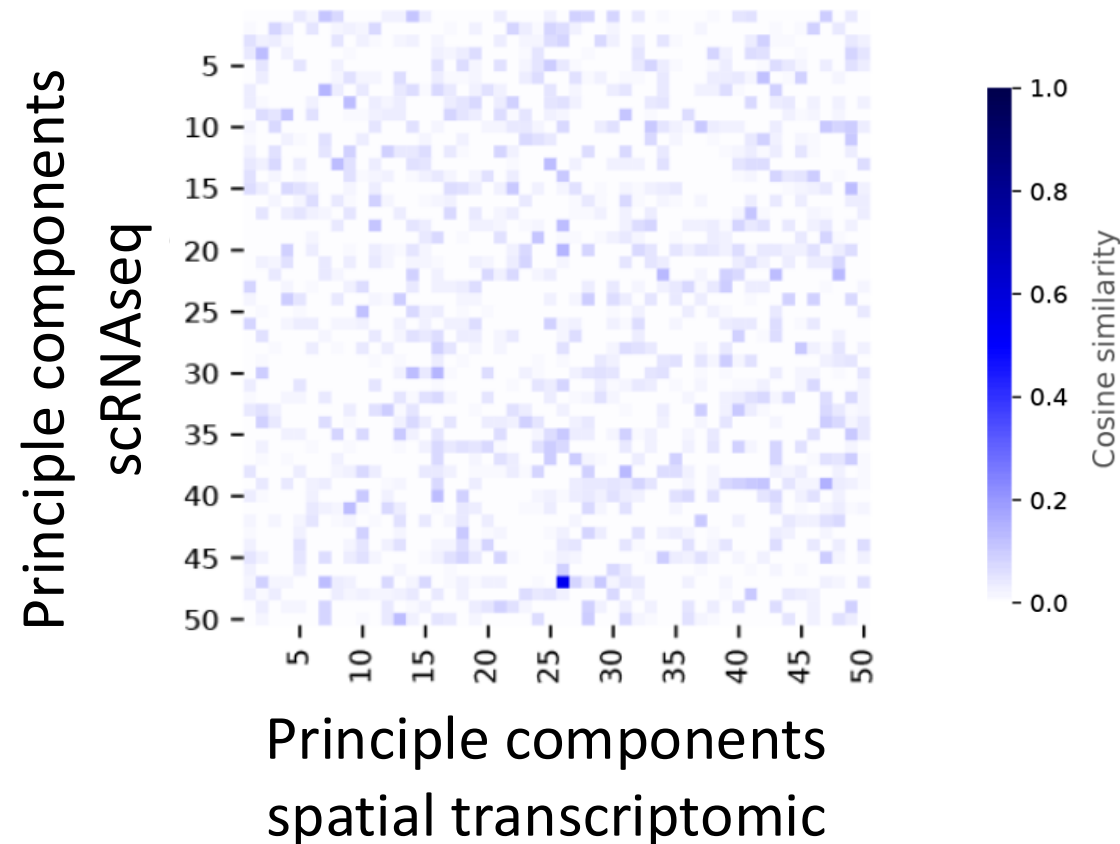
Tamim
Abdelaal

SpaGE: Spatial Gene Expression Enhancement



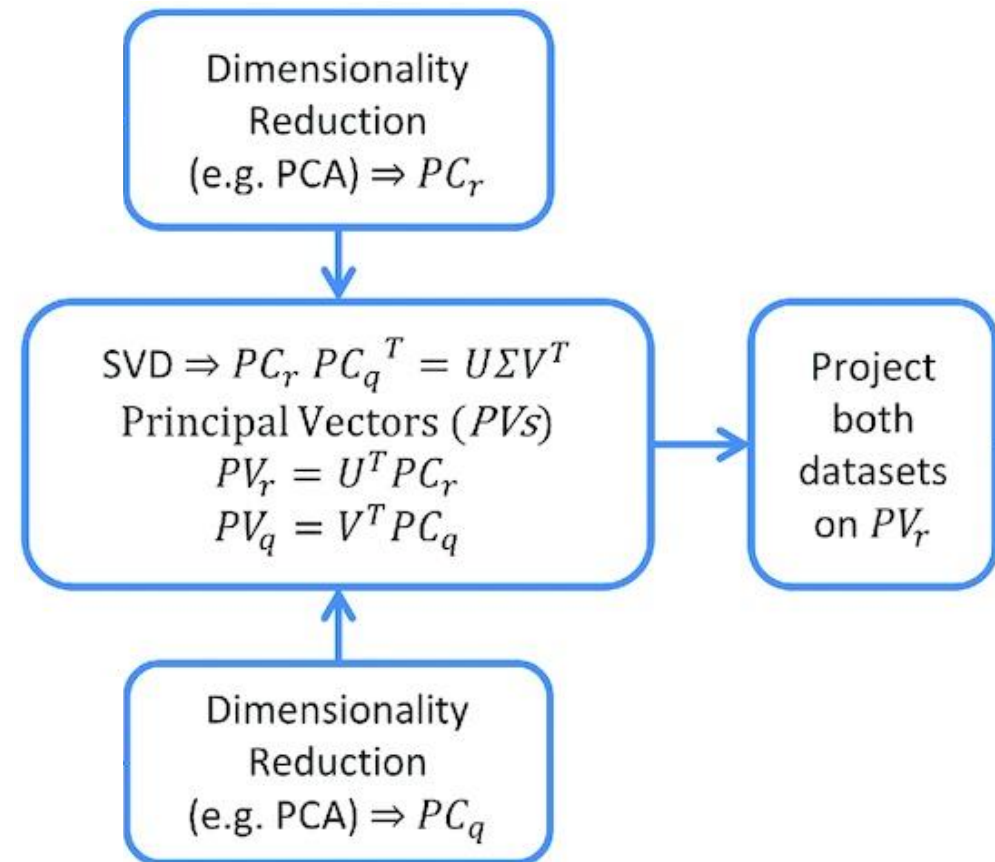
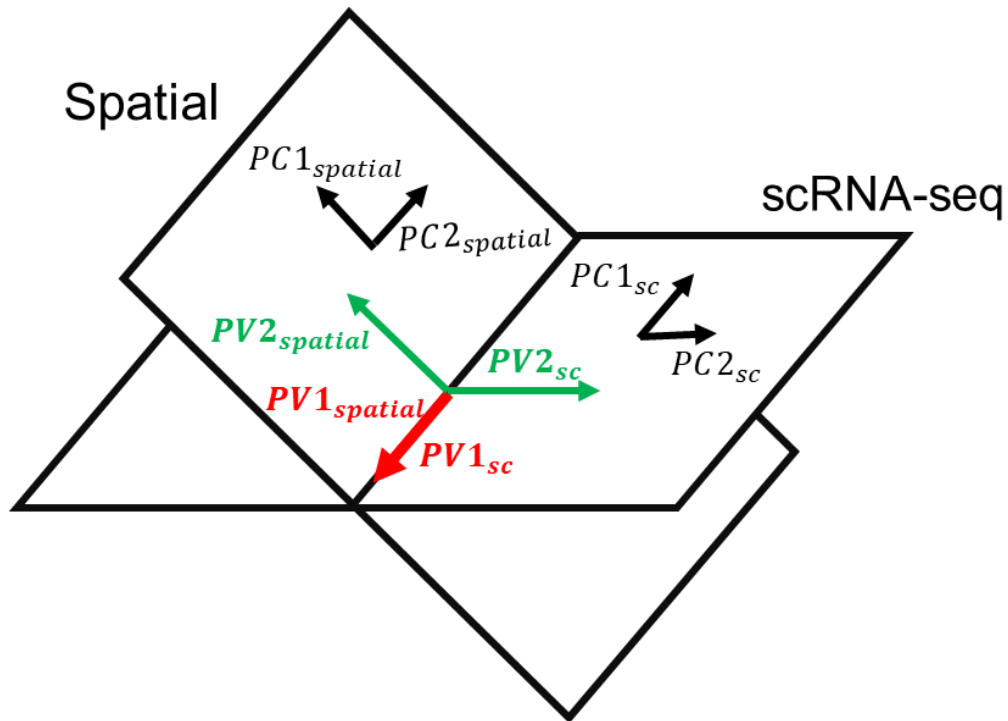
Problem: single-cell and spatial data don't align

Similarity between principal components

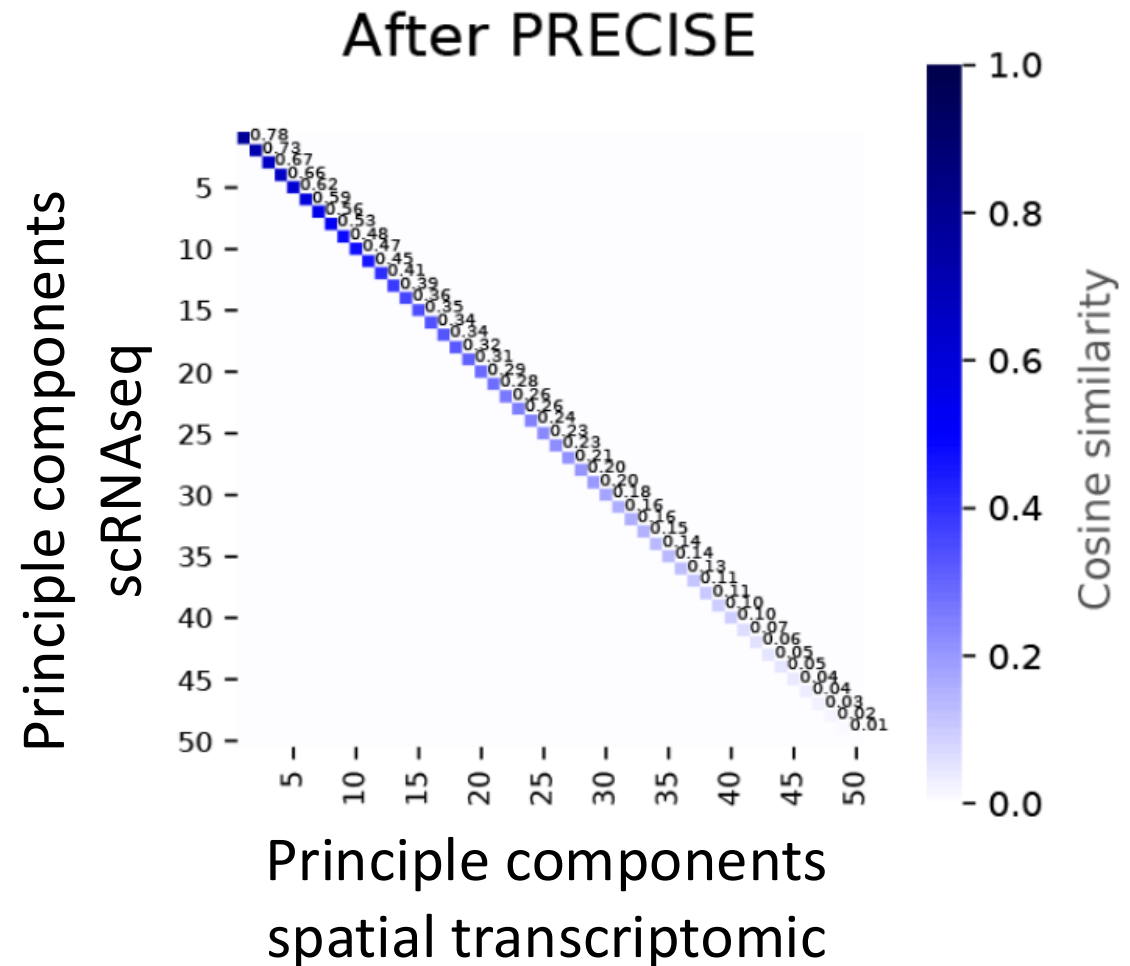
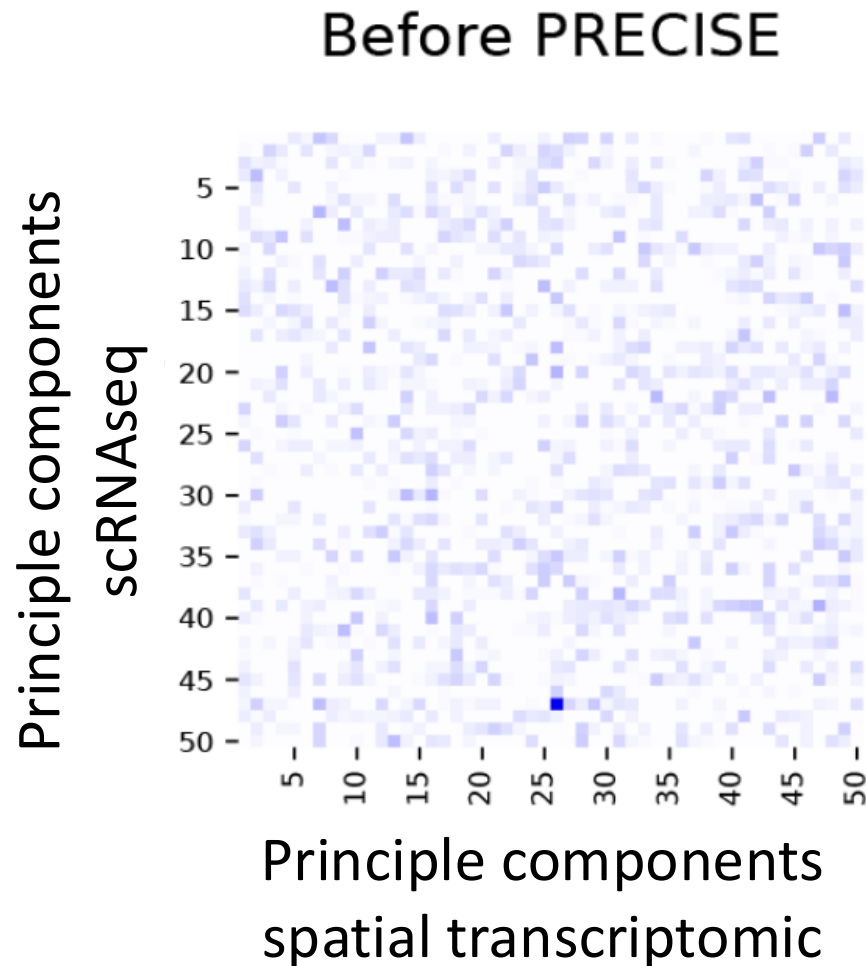


Aligning single-cell and spatial data

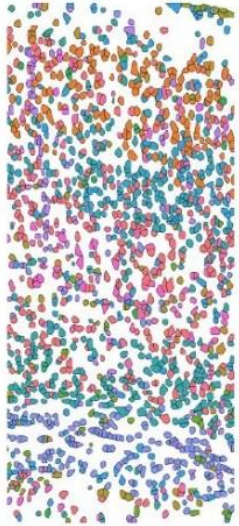
Domain Adaptation using PRECISE



Aligning single-cell and spatial data



SpaGE in primary visual cortex (VISp)

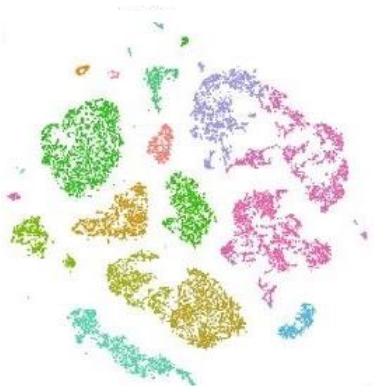


STARmap

1,549 cells

1,020 genes

Wang et al. Science 2018



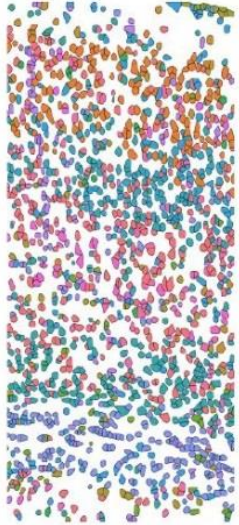
scRNA-seq

14,249 cells

34,617 transcripts

Tasic et al. Nature 2018

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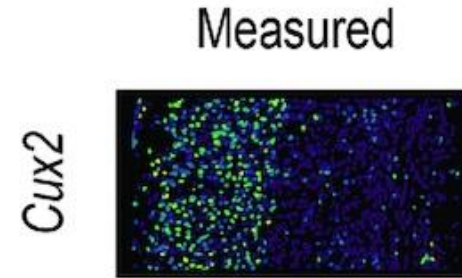


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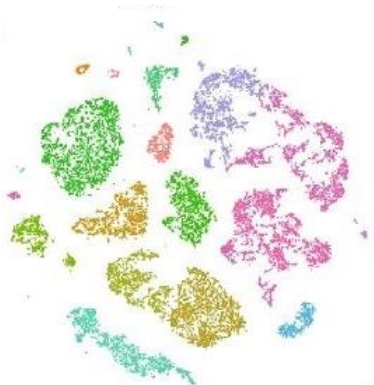


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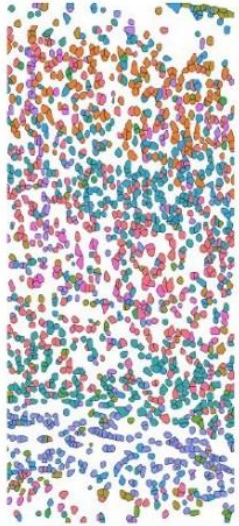
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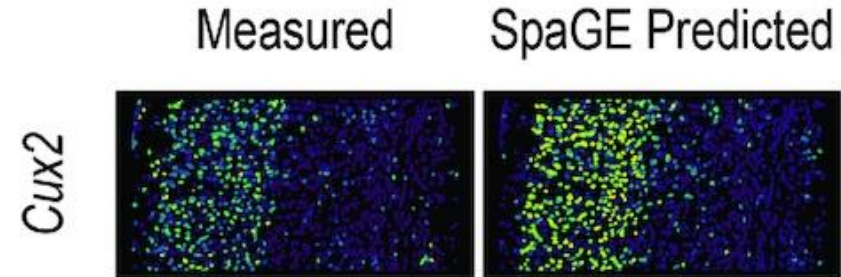


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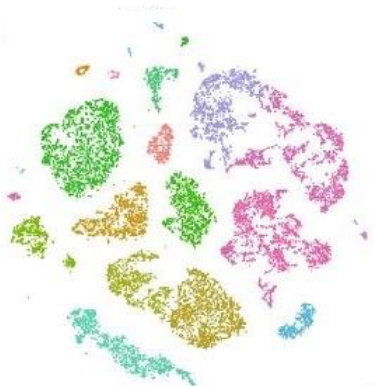


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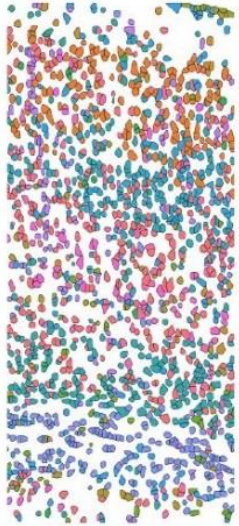
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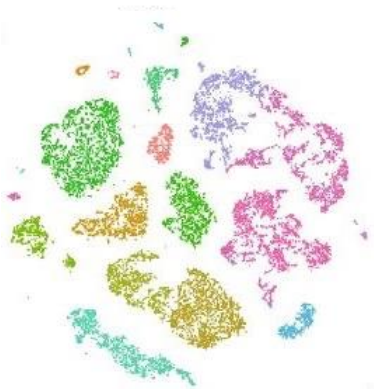


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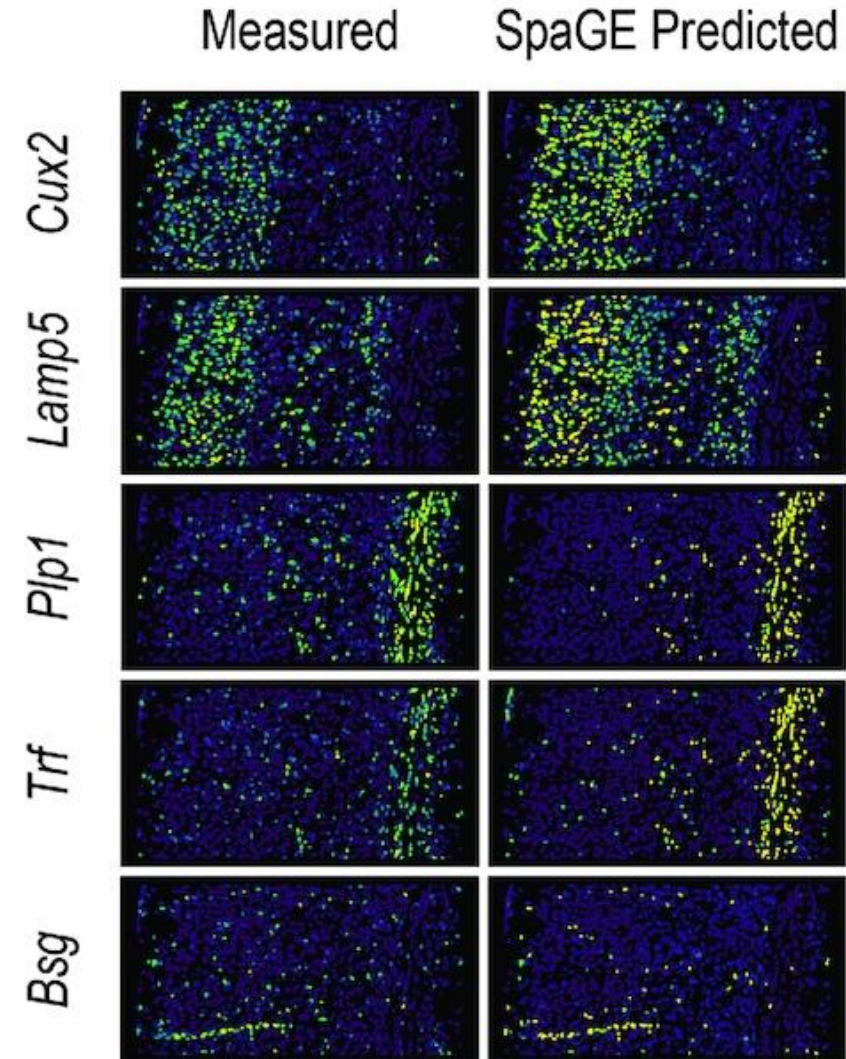


scRNA-seq

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34,617 transcripts

Tasic et al. Nature 2018



How do all these methods compare to each other?

nature communications



Article

<https://doi.org/10.1038/s41467-023-37168-7>

A comprehensive benchmarking with practical guidelines for cellular deconvolution of spatial transcriptomics

Received: 30 September 2022

Accepted: 3 March 2023

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Check for updates

Haoyang Li^{1,2,6}, Juexiao Zhou^{1,2,6}, Zhongxiao Li^{1,2}, Siyuan Chen^{1,2}, Xingyu Liao^{1,2}, Bin Zhang^{1,2}, Ruochi Zhang³, Yu Wang³, Shiwei Sun^{4,5} & Xin Gao^{1,2} ✉

Spatial transcriptomics technologies are used to profile transcriptomes while preserving spatial information, which enables high-resolution characterization of transcriptional patterns and reconstruction of tissue architecture. Due to the existence of low-resolution spots in recent spatial transcriptomics technologies, uncovering cellular heterogeneity is crucial for disentangling the spatial patterns of cell types, and many related methods have been proposed. Here, we benchmark 18 existing methods resolving a cellular deconvolution task with 50 real-world and simulated datasets by evaluating the accuracy, robustness, and usability of the methods. We compare these methods comprehensively using different metrics, resolutions, spatial transcriptomics technologies, spot numbers, and gene numbers. **In terms of performance, CARD, Cell2location, and Tangram are the best methods for conducting the cellular deconvolution task.** To refine our comparative results, we provide decision-tree-style guidelines and recommendations for method selection and their additional features, which will help users easily choose the best method for fulfilling their concerns.

Li*, Zhou* ..., Gao(2022)

ANALYSIS

<https://doi.org/10.1038/s41592-022-01480-9>

nature | methods

Check for updates

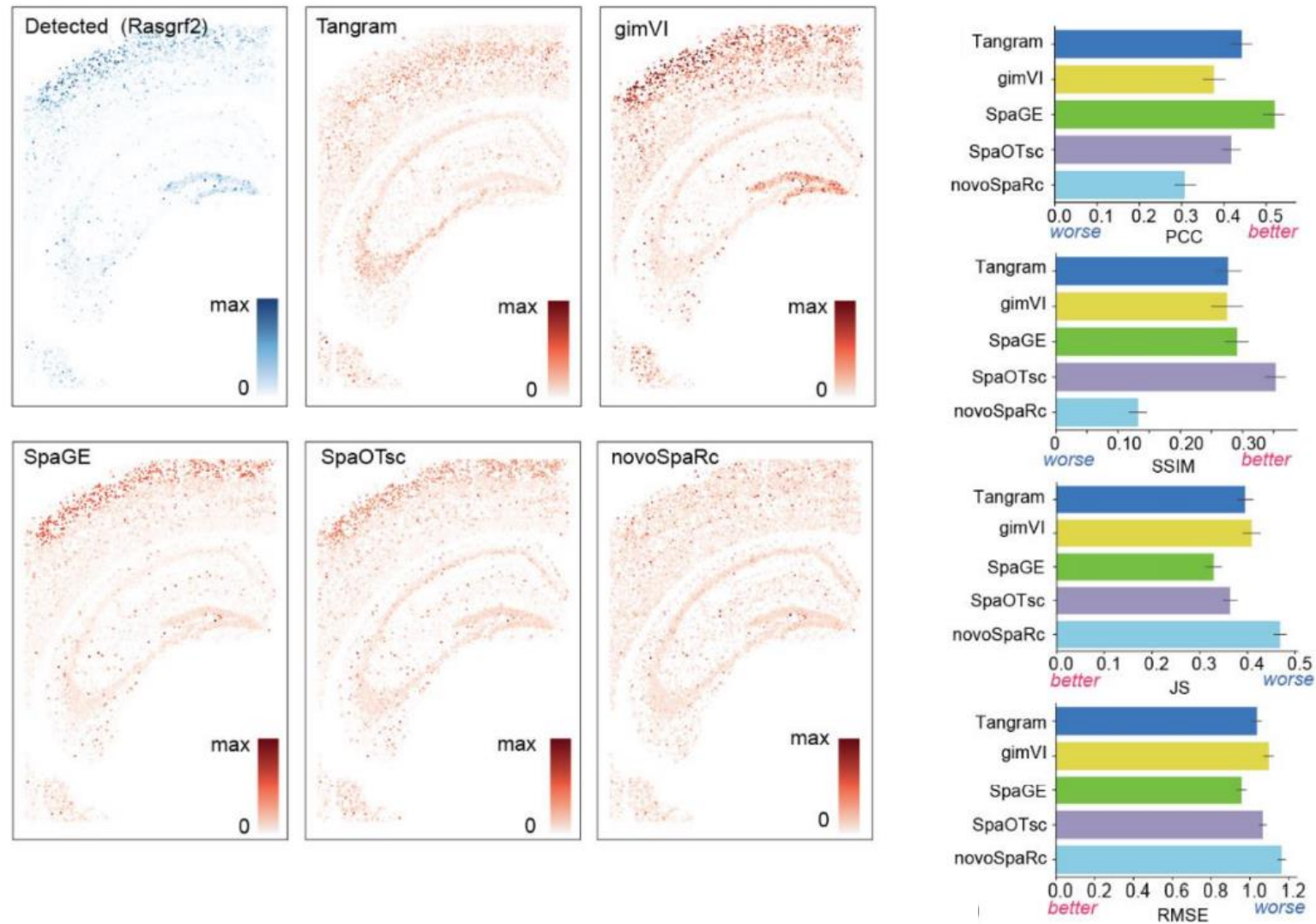
Benchmarking spatial and single-cell transcriptomics integration methods for transcript distribution prediction and cell type deconvolution

Bin Li^{1,7}, Wen Zhang^{1,2,7}, Chuang Guo^{1,7}, Hao Xu^{1,2}, Longfei Li³, Minghao Fang³, Yinlei Hu⁴, Xinye Zhang³, Xinfeng Yao¹, Meifang Tang¹, Ke Liu¹, Xuotong Zhao⁵, Jun Lin^{1,2}, Linzhao Cheng³, Falai Chen⁴, Tian Xue³ and Kun Qu^{1,2,6} ✉

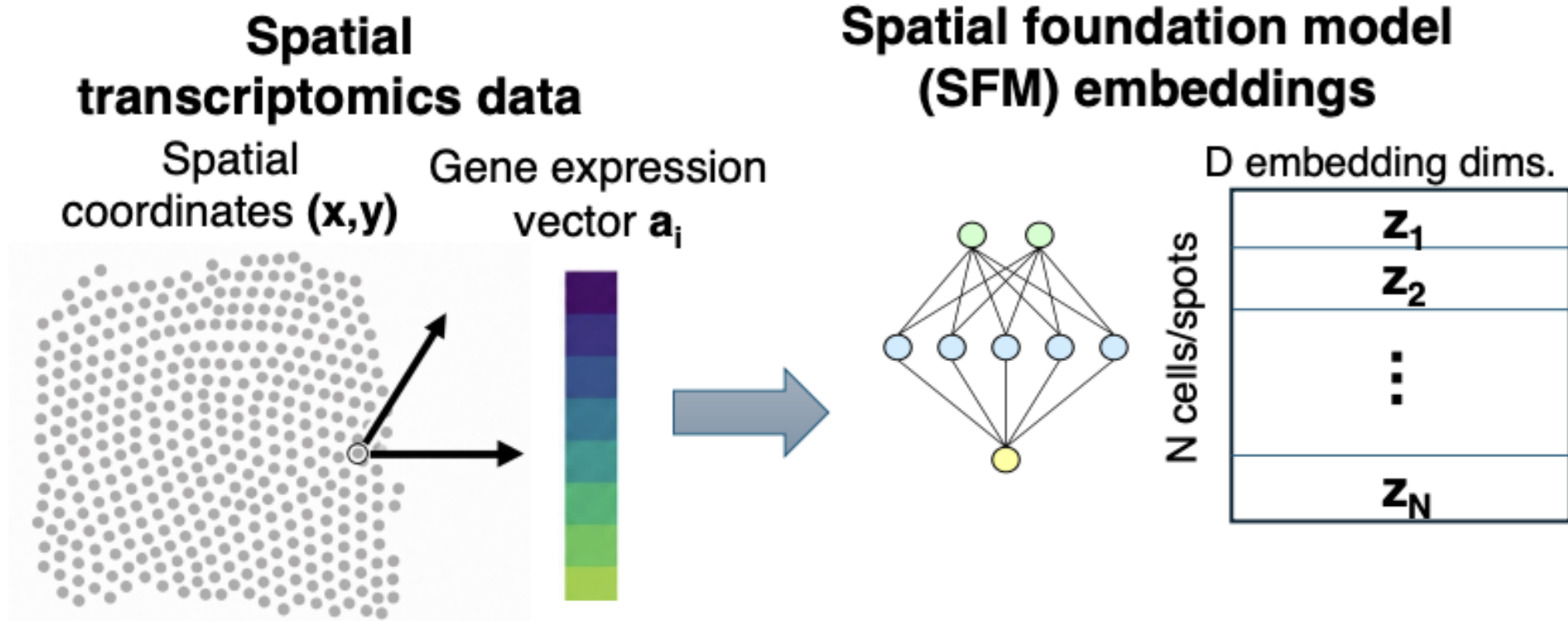
Spatial transcriptomics approaches have substantially advanced our capacity to detect the spatial distribution of RNA transcripts in tissues, yet it remains challenging to characterize whole-transcriptome-level data for single cells in space. Addressing this need, researchers have developed integration methods to combine spatial transcriptomic data with single-cell RNA-seq data to predict the spatial distribution of undetected transcripts and/or perform cell type deconvolution of spots in histological sections. However, to date, no independent studies have comparatively analyzed these integration methods to benchmark their performance. Here we present benchmarking of 16 integration methods using 45 paired datasets (comprising both spatial transcriptomics and scRNA-seq data) and 32 simulated datasets. **We found that Tangram, gimVI, and SpaGE outperformed other integration methods for predicting the spatial distribution of RNA transcripts, whereas Cell2location, SpatialDWLS, and RCTD are the top-performing methods for the cell type deconvolution of spots.** We provide a benchmark pipeline to help researchers select optimal integration methods to process their datasets.

Li*, Guo*, ..., Qu (2022)

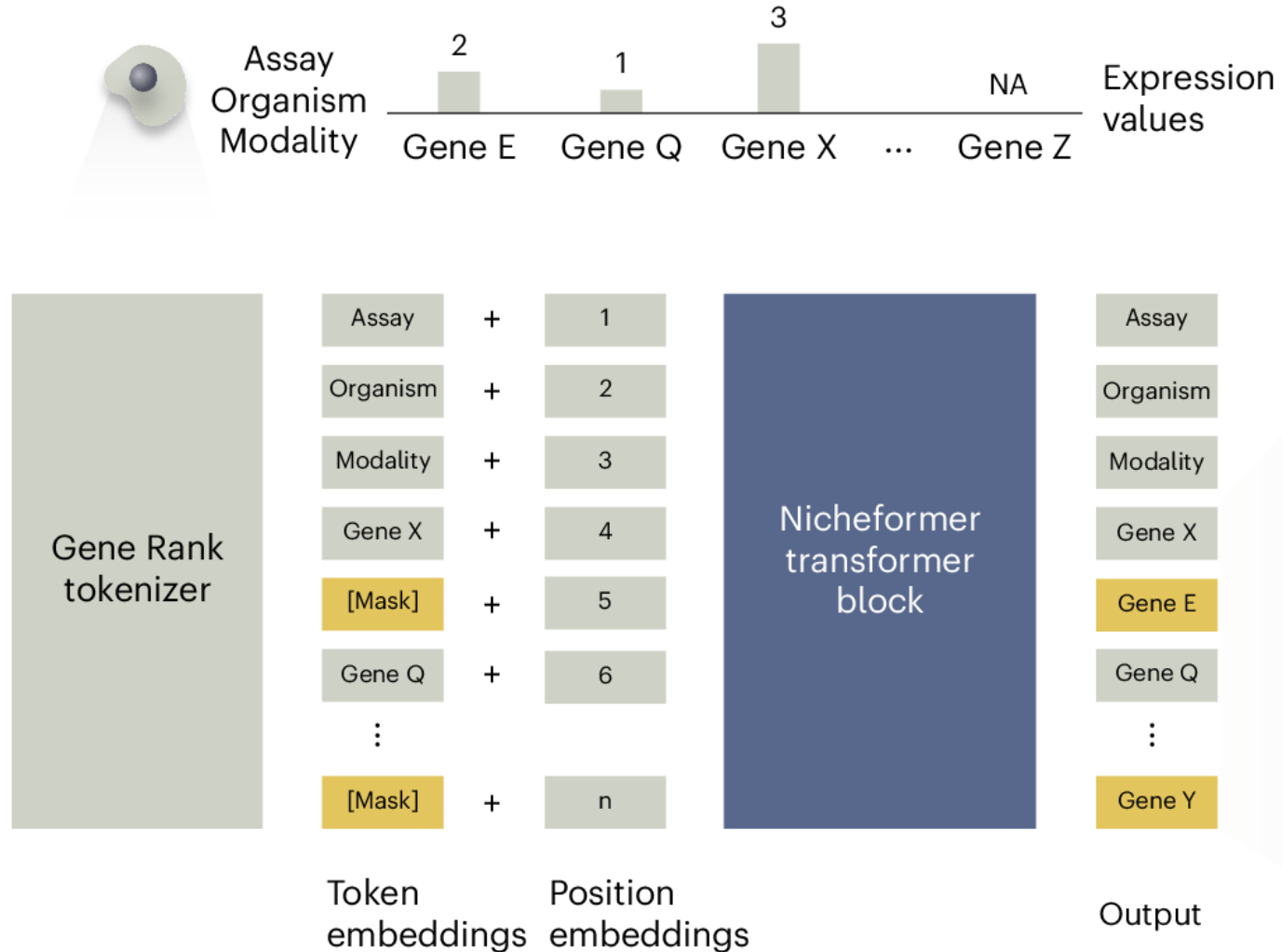
Imputation performance on Xenium data



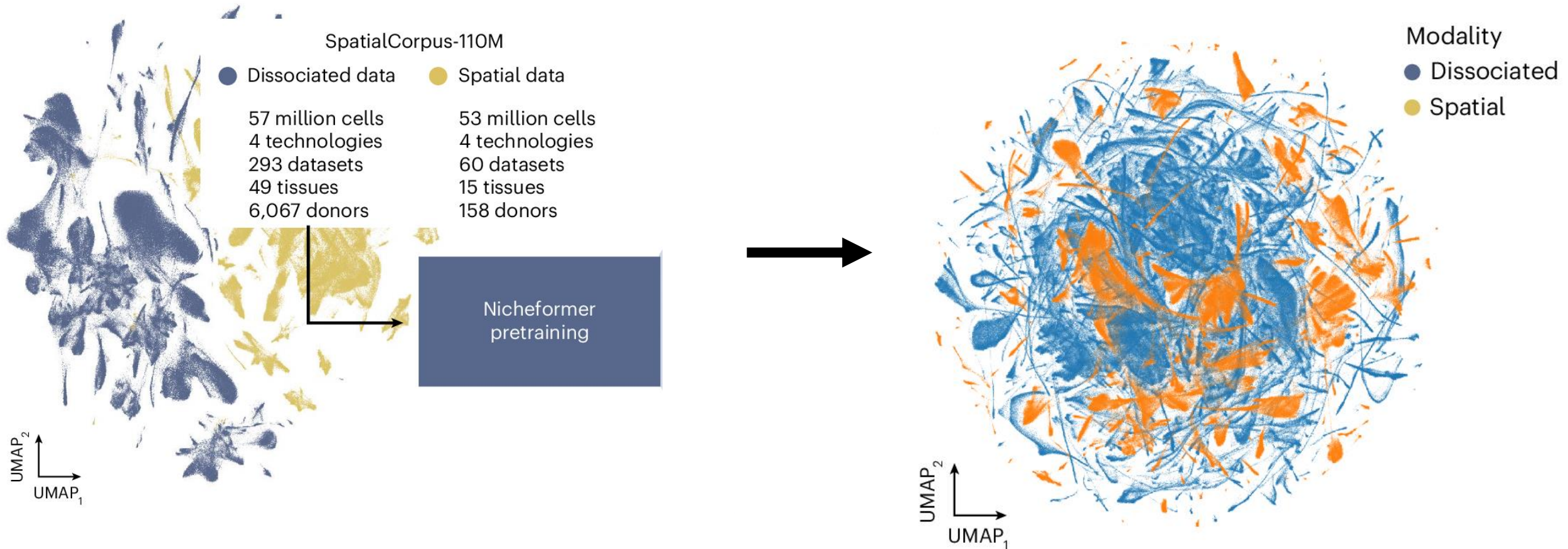
Spatial/single-cell foundation models



NicheFormer



NicheFormer

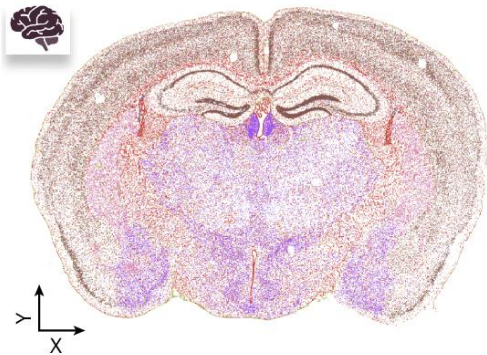


Label transfer using NicheFormer

Spatial cell-type label transfer

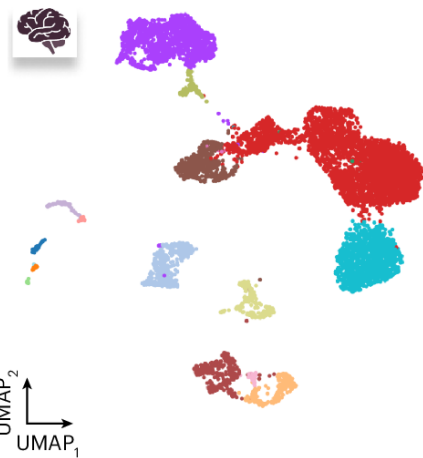


MERFISH mouse brain
Niche label



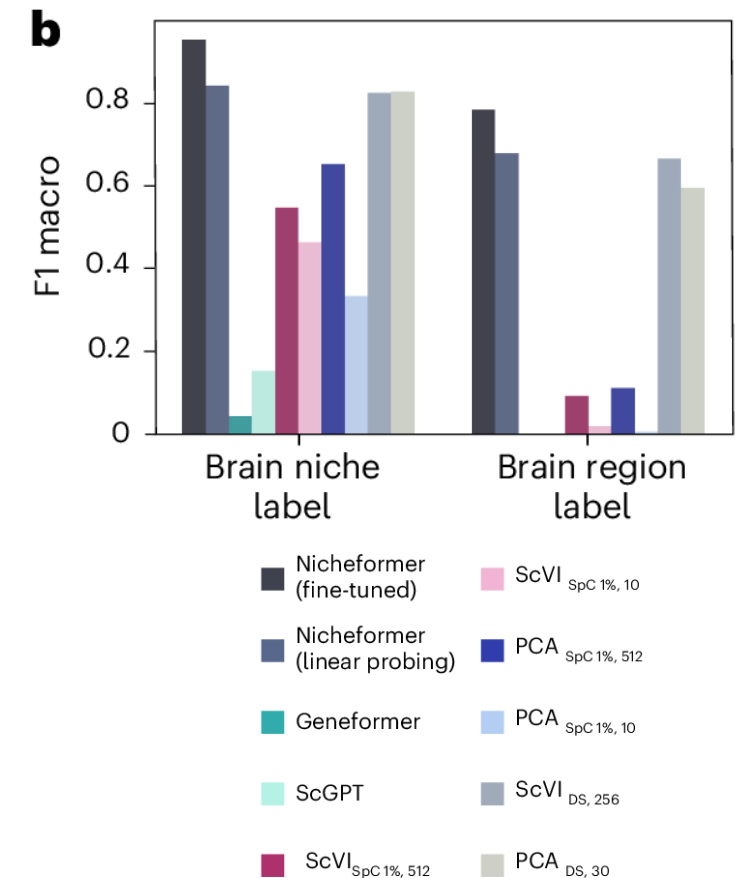
- CBX-MOB- other neuronal
- Immune
- LQ
- Neuroglial
- PAL-sAMY-TH- HY-MB-HB neuronal
- Pallium glutamatergic
- Subpallium GABAergic
- Vascular

Dissociated scRNA-seq
Mouse brain – Motorcortex



Cell types

- Astro
- Endo
- L2/3 IT
- L5 ET
- L5 IT
- L5/6 NP
- L6b
- L6 CT
- L6 IT
- L6 IT Car3
- Lamp5
- Microglia
- Oligo
- OPC
- Pvalb
- Snai3
- Sst
- Vip
- VLMC



Summary

- Integrating single-cell RNA-seq with spatial data can be used for deconvolution, localization, imputation of unmeasured genes, and annotation transfer
- Main approaches are: probabilistic mapping (e.g. Tangram), joint embedding (e.g. SpaGE), and probabilistic modelling (e.g. cell2location)
- Single-cell/spatial foundation models can also be used to learn a joint embedding of both data types

Thank You!

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